

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750–2024

Time Span

35+ Bt

Annual Emissions

429 ppm

Atmospheric CO₂



Atmospheric CO₂ Reaches 429 ppm

50% above pre-industrial levels — a change that took millennia in the past now occurs in decades

429

PARTS PER MILLION (FEB 2026)

Acceleration Context

That natural increase at the end of the last ice age took **thousands of years**. The current surge has occurred in roughly **two centuries** — driven primarily by burning coal, oil, and natural gas.

Measurement: NOAA Mauna Loa Observatory

China Now Emits Over 12 Billion Tonnes

More than double the United States — over one-quarter of the global total

COUNTRY/REGION	ANNUAL CO ₂ (BT)	SHARE	TREND
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising fast
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Stable
France	~0.3	~0.8%	Declining

Data visualization: figure_3.jpg

Key Observation

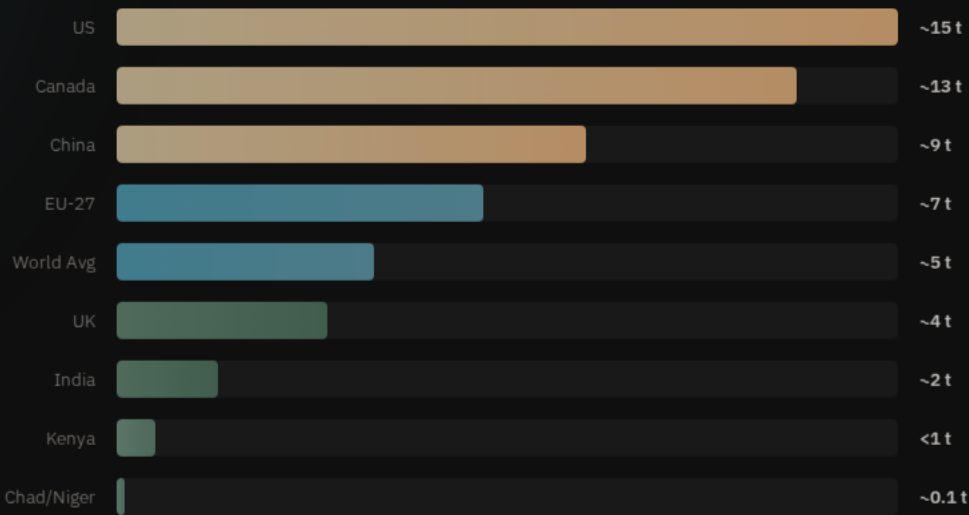
The US peaked near **6 billion tonnes** around 2005 and has since declined to roughly **5 billion tonnes**.

India has risen rapidly to approximately **3 billion tonnes** — now the third-largest and climbing.

Note: Europe's share has fallen substantially from its historical dominance.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad

A 150x disparity — the average American emits in under two days what the average person in Chad emits in a year



Extreme Inequality

The gap between highest and lowest per capita emitters is approximately **150-fold**.

Among large populous countries, the US, Australia, and Canada have per capita emissions roughly **3x the global average**.

Note: France (~4 t) and UK (~4 t) have much lower per capita emissions than Germany or Netherlands despite similar living standards — due to energy mix.

Per capita chart: [figure_1.jpg](#)

2024 Snapshot: Diverging Trajectories

US and Europe cut emissions while Russia, Southeast Asia, and parts of Africa increase

Declining (-1% to -5%)

- United States
- Much of Europe

Shown in blue on the map — structural shifts toward cleaner energy and efficiency measures

Rising (+2% to +10%+)

- Russia
- Parts of Southeast Asia
- Several African nations
- Some South American countries (>10%)

Shown in orange/red on the map — continued reliance on fossil fuels for development

Map reference: figure_4.jpg

Energy Mix Explains Why France Emits Far Less Than Germany

Countries with similar living standards can have very different per capita emissions depending on energy sources

France & UK		Germany & Netherlands	
Per Capita CO ₂	~4 t	Per Capita CO ₂	~8–10 t
Nuclear Share	High (~70% France)	Nuclear Share	Low (Germany phasing out)
Renewables	Substantial	Renewables	Growing but not dominant
Fossil Fuel Dependence	Low	Fossil Fuel Dependence	High (coal + gas)

